

Smarter technology for all

Servicing the ThinkSystem SR685a V3

ES72656

April 2025

Lenovo

Prerequisites

- [ES42430 – AMD EPYC processor architecture for ThinkSystem V3 servers](#)
- [ES51757B – Introducing ThinkSystem tools](#)
- [ES52374 – ThinkSystem tools for the ThinkSystem V3 platform](#)
- [ES41759C – ThinkSystem problem determination](#)
- [ES51780C – Servicing the ThinkSystem storage controllers](#)

Objectives

After completing the course, you will be able to:

- Describe the ThinkSystem SR685a V3 and its components
- List the SR685a V3 specifications
- Describe the SR685a V3 configurations and block diagrams
- Describe the SR685a V3 management tools
- Describe the problem determination steps and explain how to troubleshoot issues with the SR685a V3

Product overview

Product description and front, rear, and inside views

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SR685a V3 product overview

The ThinkSystem SR685a V3 is an 8U, two-socket, GPU-rich rack server. The SR685a V3 is designed for demanding AI computing or HPC workloads.

The SR685a V3 supports two 4th Gen AMD EPYC CPUs, and eight NVIDIA or AMD OAM (open project computing accelerator module) GPUs. Supported GPUs include the NVIDIA H100 and H200, and the AMD MI300X OAM GPUs.



Specifications

Features	Specifications
Form factor	8U rack server
Processors	Two 4 th Gen AMD EPYC processors, TDP up to 400 W
Memory	Up to 3 TB with 24 DDR5 4800 MHz DIMMs
Internal storage	• 16 2.5-inch U.2 or U.3 NVMe drives at the front • Two M.2 NVMe drives inside the chassis as a boot drive device
I/O devices	• Front: Eight NVIDIA ConnectX-7 PCIe 5.0 FHHL network adapters • Rear: Two NVIDIA ConnectX-7 PCIe 5.0 FHHL network adapters or One NVIDIA ConnectX-7 PCIe 5.0 FHHL network adapter and one OCP adapter
PSUs	Eight hot-swap 2600 W CFFv4 PSUs, support for N+N redundancy
Cooling	• Five hot-swap fans at the front for the compute shuttle • Ten hot-swap fans at the rear for the GPU shuttle Support for N+1 redundancy across all 15 fans
GPUs	Support for eight high-performance OAM GPUs • Eight AMD Instinct MI300X GPUs with AMD Infinity Fabric interconnects at 800 GBps • Eight NVIDIA HGX H100 or H200 GPUs with NVIDIA NVLink interconnects at 900 GBps
Management tools	XClarity series tools: • XClarity Controller (XCC) • XClarity Provisioning Manager (LXPM) • XClarity Essentials (LXCE)

Front view

Front fans

Five front hot-swap fans

Integrated diagnostics panel

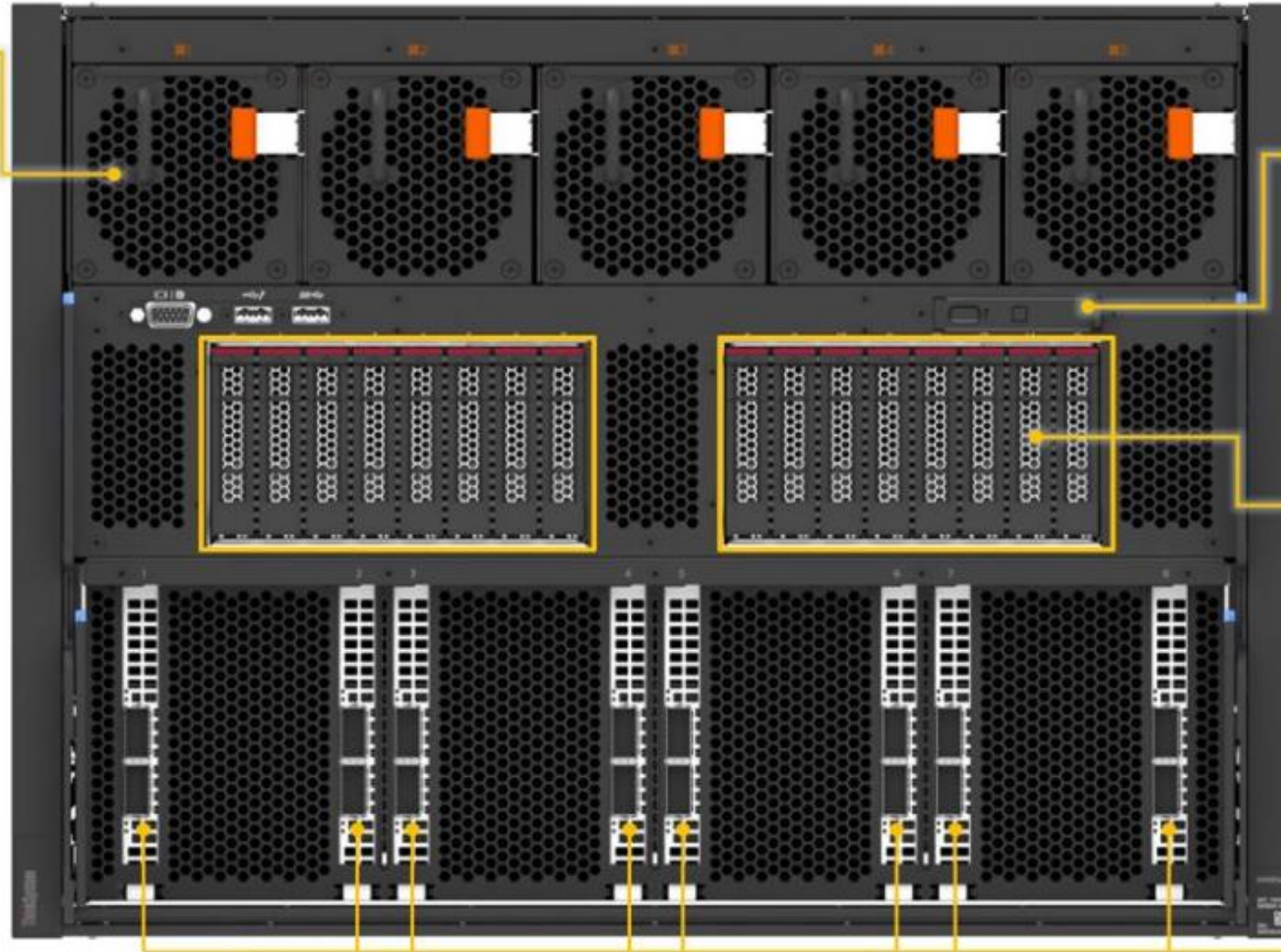
Includes system LEDs and power button

Front drives

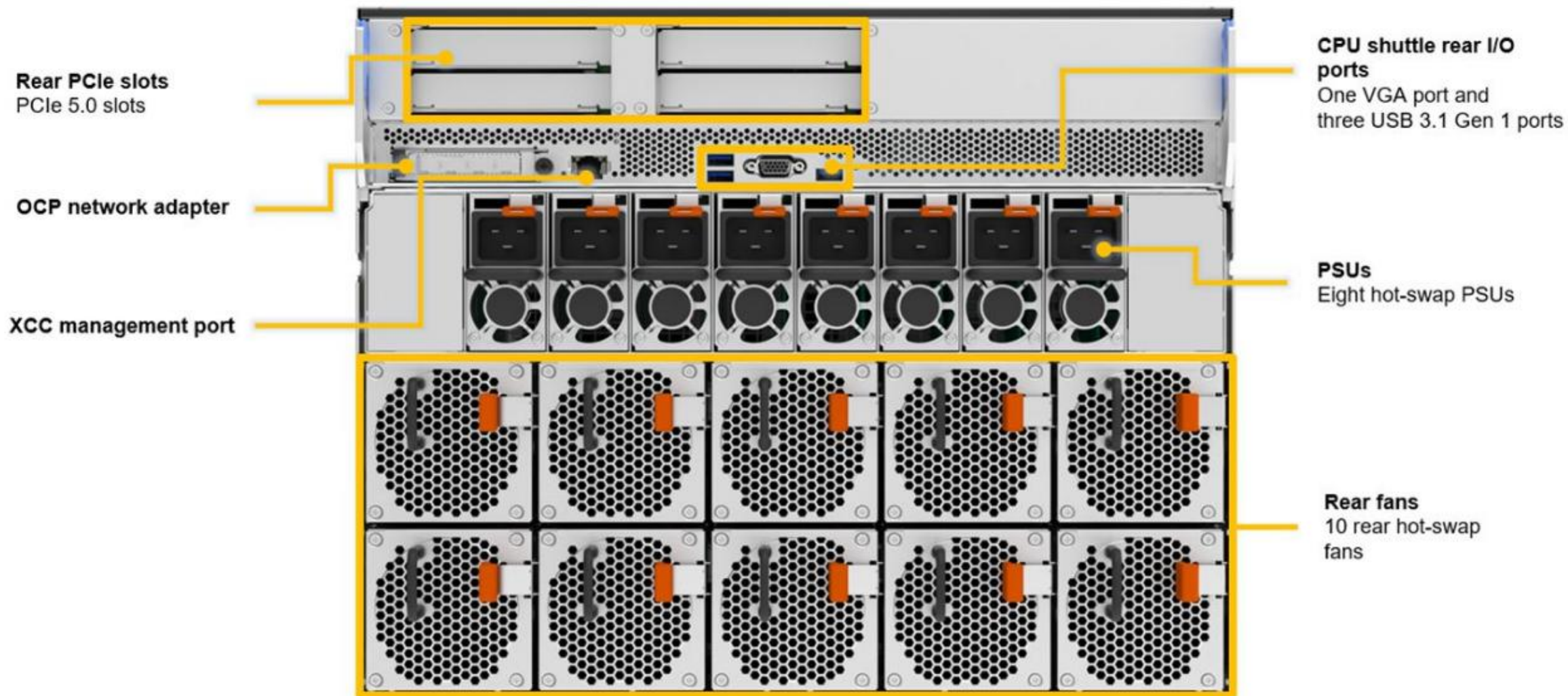
16 2.5-inch hot-swap NVMe drives

PCIe adapters

Eight PCIe 5.0 x16 FHFL adapters



Rear view



Inside views

Click the buttons to see inside views of the SR685a V3.



Exploded view

Inside view without
chassis

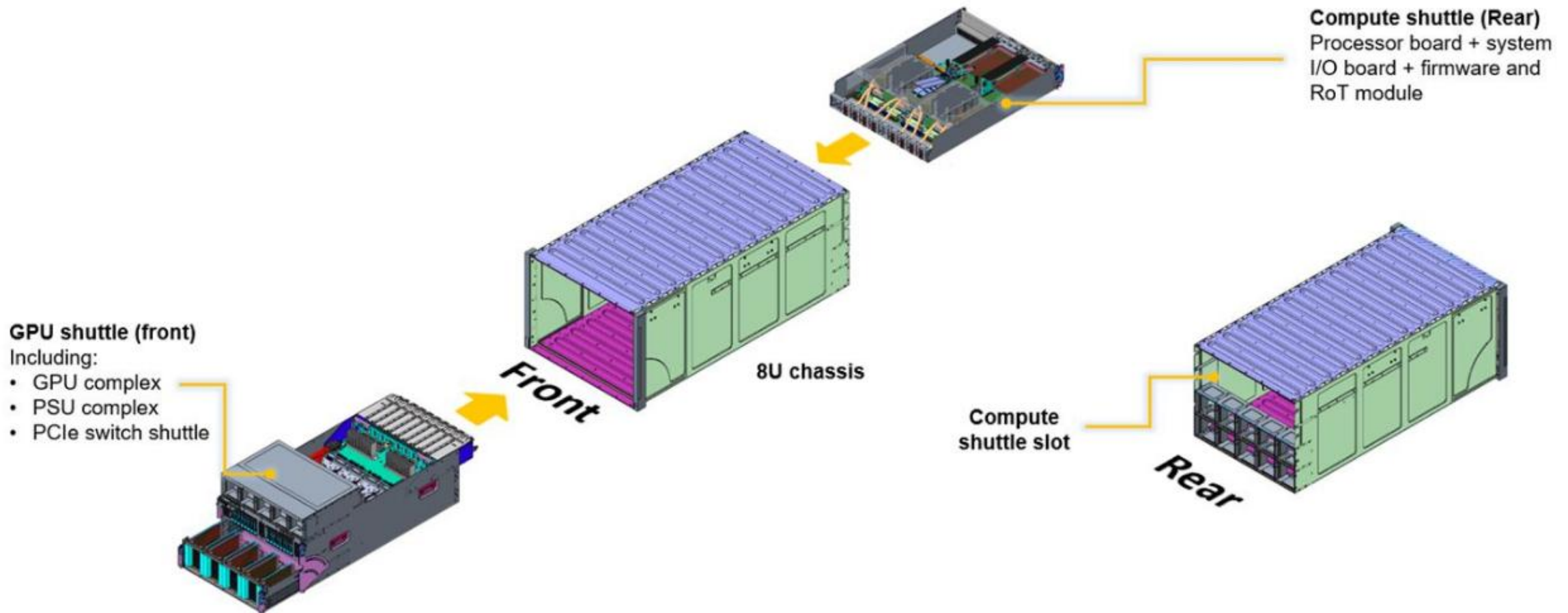
Side view

Compute shuttle

GPU shuttle

Inside views

Click the buttons to see inside views of the SR685a V3.



Exploded view

Inside view without chassis

Side view

Compute shuttle

GPU shuttle

Inside views

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Front fans

Five front hot-swap fans

Hot-swap drives

16 2.5-inch NVMe drives

Switch board

PCIe lane signal connections for internal GPUs to the system board assembly

System board assembly

Processor board + system I/O board + firmware and RoT module

PSU complex

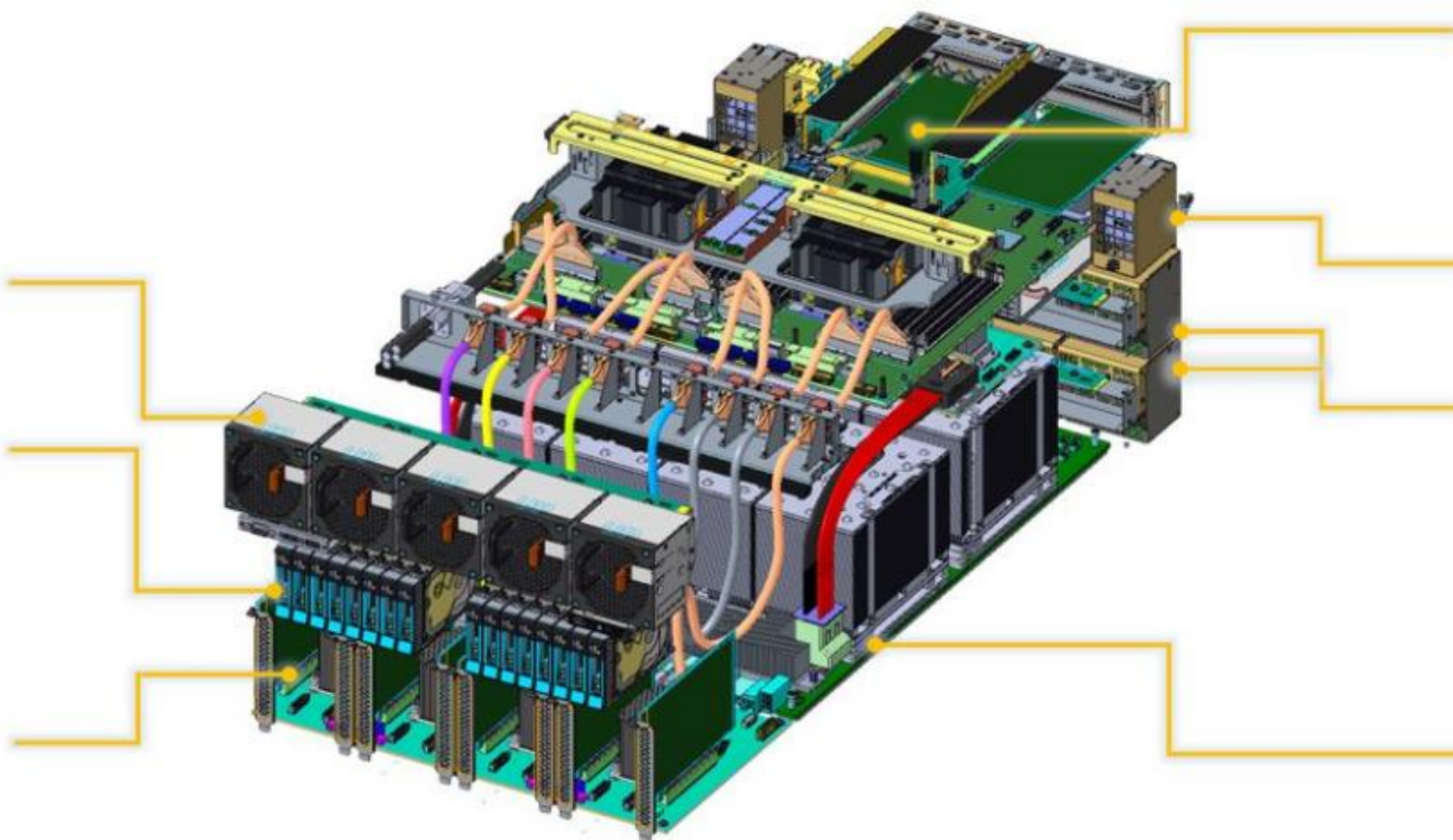
Eight hot-swap PSUs

Rear fans

10 rear hot-swap fans

GPU complex

Includes eight NVIDIA or AMD OAM GPUs, and a GPU baseboard



Exploded view

Inside view without chassis

Side view

Compute shuttle

GPU shuttle

Inside views

Click the buttons to see inside views of the SR685a V3.



Front fans
Five front hot-swap fans

Integrated diagnostics panel
Includes system LEDs and power button

Hot-swap drives
16 2.5-inch NVMe drives

Switch board
PCIe lane signal connections for internal GPUs to the system board assembly

Switch cables harness
CPU shuttle and GPU shuttle cable connections

System board assembly
System board assembly

PSU complex
Eight hot-swap PSUs and power distribution board

GPU complex
Includes eight NVIDIA or AMD OAM GPUs, and a GPU baseboard

Rear fans
10 rear hot-swap fans

Exploded view

Inside view without chassis

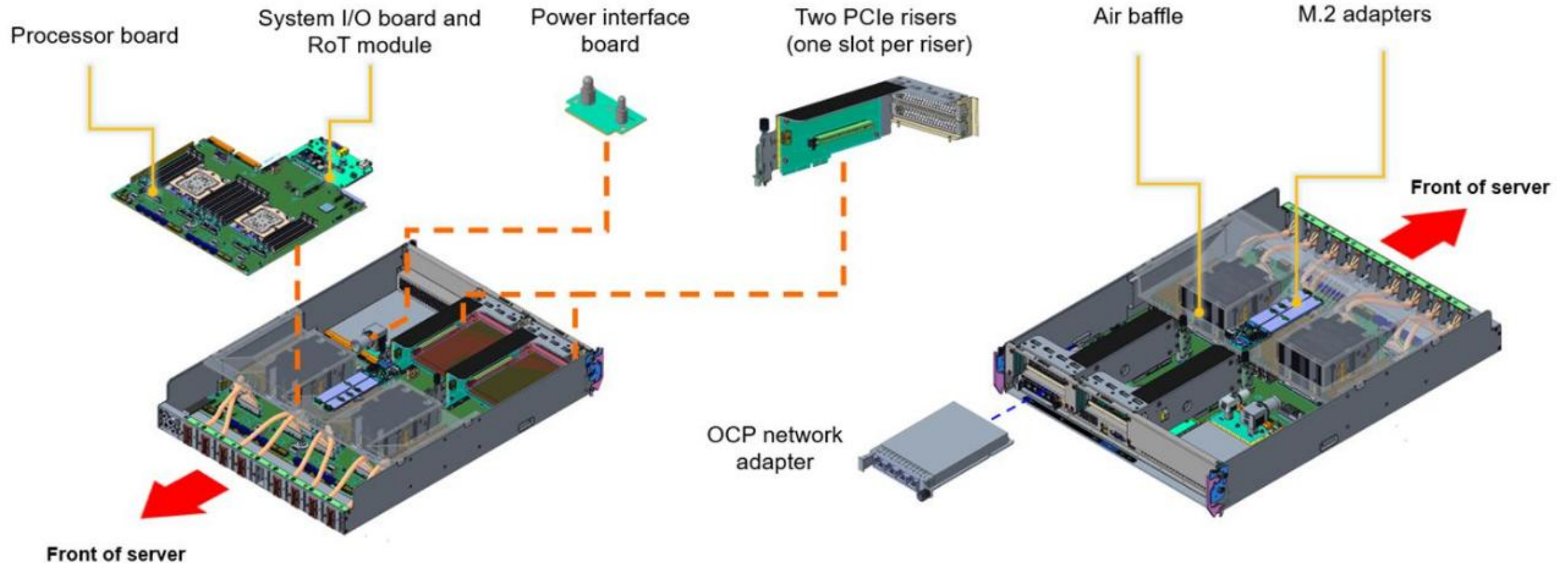
Side view

Compute shuttle

GPU shuttle

Inside views

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Exploded view

Inside view without chassis

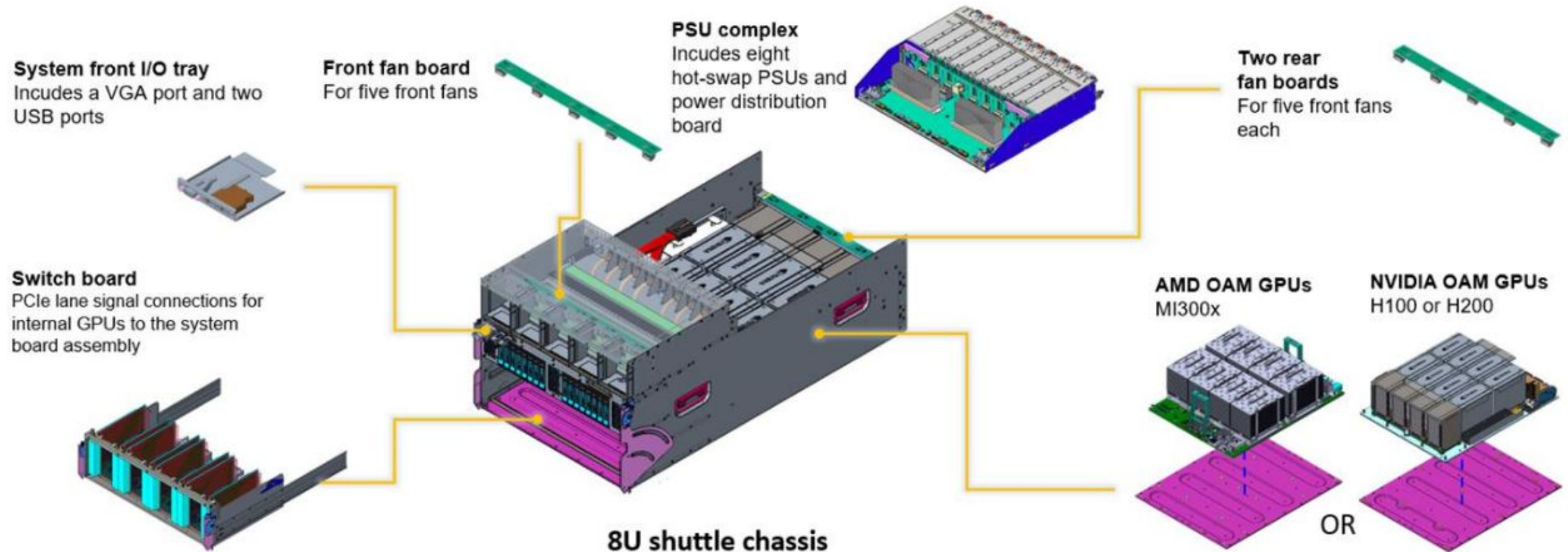
Side view

Compute shuttle

GPU shuttle

Inside views

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Exploded view

Inside view without chassis

Side view

Compute shuttle

GPU shuttle